

# **PerceptEEG: Diagnosing and Bridging the EEG–Visual Representation Gap for Zero-Shot EEG-to-Image Retrieval**

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## Abstract

EEG-to-image retrieval identifies viewed images from visually evoked brain activity, yet existing methods commonly align noisy EEG signals with fixed, semantically rich visual targets without first assessing their compatibility with EEG. This dissertation investigates this representation mismatch through a controlled diagnostic study of what visual information is recoverable, when and where it is supported, and which depth of a CLIP representation provides the closest correspondence. The results reveal a substantially larger mismatch for semantic than perceptual information, identify the 100–300, ms temporal window and posterior sensors as the strongest input restriction under the adopted probing protocol, and select Block 14 as the most compatible visual depth among the evaluated layers. Guided by these findings, we propose **PerceptEEG**, comprising two components: a compact Mid-level Visual Target (MVT) that provides a perceptually compatible visual representation, and a Windowed Spatiotemporal Residual Adapter (WSRA) that enables bounded refinement of EEG embedding features. On 200-way retrieval, PerceptEEG achieves  $65.85 \pm 7.46\%$  intra-subject and  $12.85 \pm 4.76\%$  cross-subject Recall@1. It achieves state-of-the-art-level performance in the intra-subject setting and ranks second among the included methods under cross-subject leave-one-subject-out evaluation, with  $34.20 \pm 9.85\%$  Recall@5. At 25% training-concept coverage, it retains 79.4% and 90.9% of its full-data cross-subject Recall@1 and Recall@5, respectively. Staged ablation attributes the largest measured improvement to MVT, with WSRA providing an additional gain when combined with MVT. These findings highlight target compatibility as an important design factor for accurate and data-efficient EEG–image alignment, while subject-invariant transfer remains an open challenge.

## **Declaration of originality**

I hereby confirm that no portion of the work referred to in the thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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**Acknowledgments**

# 1 Introduction

## 1.1 Background

Neural decoding maps brain activity to externally interpretable representations, supporting research in assistive communication, device control, and neurorehabilitation [1], [2], while providing a quantitative tool for studying human perception [3], [4]. Within this broad area, visual neural decoding seeks to recover perceived content from stimulus-evoked responses. EEG-to-image retrieval operationalizes this objective as a ranking problem: given a visually evoked electroencephalography (EEG) signal, a model must identify the viewed image from an unseen candidate gallery. The task therefore offers both a controlled benchmark of visual representation learning and a foundational capability for neural interfaces that aim to communicate perceptual content.

Among non-invasive recording modalities, EEG offers a practical balance of low cost, portability, and millisecond temporal resolution. By comparison, functional magnetic resonance imaging (fMRI) provides finer spatial measurements but is constrained by slow hemodynamics and a fixed scanner, whereas magnetoencephalography (MEG) typically requires shielded instrumentation [3], [4], [5]. These practical advantages come with substantial modeling challenges: scalp EEG is noisy and spatially mixed, paired EEG–image data are scarce relative to visual pretraining corpora, and response distributions vary across individuals. The central question is therefore not only whether EEG can retrieve visual stimuli, but whether it can learn a visual representation that is compatible with the information present in EEG and remains robust under cross-subject transfer and limited-data training.

## 1.2 Related Work

### 1.2.1 EEG-to-Image Retrieval

EEG visual decoding has progressed from closed-set category prediction toward open-set retrieval and reconstruction. Early work learned category-discriminative EEG manifolds from visually evoked responses [6]; subsequent advances in contrastive learning and pretrained vision–language models made it possible to align neural activity with transferable visual spaces [7], [8]. NICE and ATM instantiate this paradigm by aligning EEG with CLIP features, with ATM additionally modeling channel and temporal–spatial interactions [9], [10]. These systems establish a scalable retrieval formulation, but direct alignment inherits the information content and geometry of a visual target selected in advance.

Recent methods address this limitation through encoder transfer, target adaptation, or prior-driven augmentation. EEGiT converts anatomically grouped and spatially interpolated EEG into image-like

patches to reuse a pretrained vision transformer, although the resulting representation depends on a constructed spatial layout [11]. NeuroCLIP adapts the visual branch through prompt tuning, whereas SAMGA aggregates subject-aware multi-granularity visual features [12], [13]. UBP attenuates uncertain high-frequency image content through adaptive blur, and NeuroBridge combines modality-specific transformations with bidirectional semantic alignment [14], [15]. These approaches improve retrieval, but their end-to-end evaluations do not establish whether transformed visual information is absent from EEG or merely difficult for a particular encoder to recover.

### **1.2.2 EEG Representation Learning and Decoding**

EEG encoders have evolved from general convolutional architectures toward compact, task-aware spatiotemporal models. DeepNet and ShallowNet demonstrated end-to-end temporal and spatial feature learning from raw EEG, while EEGNet introduced depth-wise and separable convolutions for parameter-efficient decoding [16], [17]. TSConv and ATM further combine temporal–spatial convolution with attention or contrastive visual supervision [9], [10]. This architectural diversity motivates evaluating multiple frozen encoders in the diagnostic study; downstream accuracy alone, however, cannot identify which visual factors each representation makes accessible.

Linear probing and representational similarity analysis provide complementary tools for examining learned representations. Linear probes estimate which targets are accessible through a restricted readout [18], time-resolved multivariate decoding characterizes when task information supports prediction [19], and representational similarity analysis compares relational geometry across brains and models [20]. Our diagnostic protocol combines factor-specific linear probes with controlled temporal and channel restrictions and a comparison across visual-encoder depth. This combination measures task-specific recoverability; it does not estimate information-theoretic capacity or perform neural source localization.

### **1.2.3 Neural and Visual Representation Alignment**

Progress in pretrained vision and generative models has made the choice of neural alignment target increasingly consequential. In fMRI, latent diffusion models can reconstruct images by mapping spatially resolved neural activity to pretrained generative components [21], while human-aligned image encoders improve decoding across EEG and MEG architectures [22]. VINDEK further shows that compact visual features can outperform denser alternatives in fMRI decoding [23]. These findings suggest that richer supervision is not automatically more compatible with recorded brain activity, but differences in spatial resolution, temporal resolution, and noise prevent their direct transfer to scalp

EEG.

Brain–model comparisons provide a second basis for questioning final-layer supervision. The progression of cortical visual processing shows an ordered correspondence with neural-network depth, yet stronger task performance does not necessarily imply a more brain-like hierarchy [3], [24]. Residual adapters likewise provide a compact mechanism for introducing task- or domain-specific corrections into pretrained representations [25], but generic adaptation does not determine which EEG intervals, sensors, or visual features should guide the correction. Together, these lines of work motivate an EEG-specific diagnosis of representation content and correspondence before selecting the visual target and sensor–time priors.

### 1.3 Research Gap and Motivation

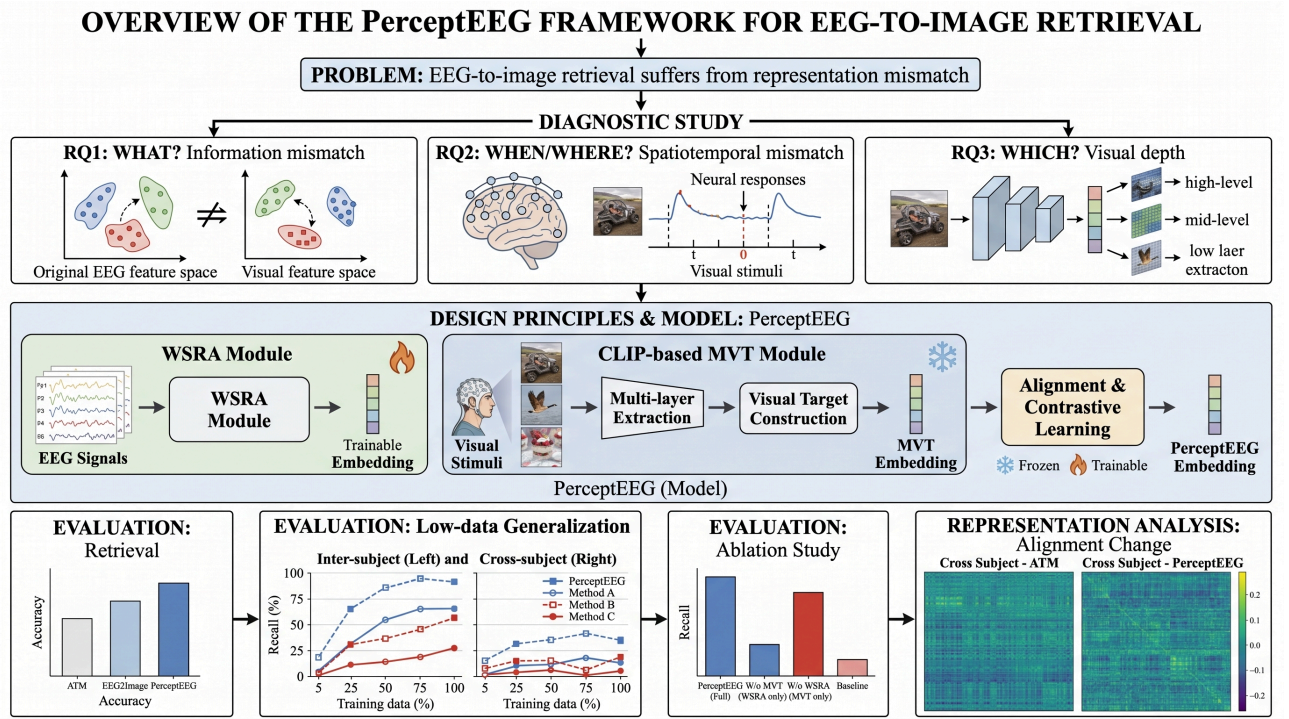
Existing work often focuses on improving the alignment pipeline without systematically characterizing the representation mismatch it is intended to address. To address this gap, we formulate three diagnostic research questions and develop a unified evaluation framework to characterize visual information in EEG, its spatiotemporal distribution, and its alignment with visual representation depth:

- RQ1.** What visual information is linearly recoverable from pretrained EEG representations, and how does it compare with matched image representations?
- RQ2.** Which temporal intervals and channel groups support visual information recovery from EEG signals?
- RQ3.** Which level of the CLIP visual hierarchy corresponds most strongly to temporally restricted EEG signals?

The diagnostic study suggests a structured rather than uniform mismatch. Perceptual information appears to be more closely matched across EEG and image representations than semantic information, while an early-to-intermediate post-stimulus interval and posterior channel restriction yield stronger measured input support. An intermediate CLIP representation also appears more compatible as a target than the sampled early and final representations. These observations provide high-level design priors for the proposed PerceptEEG framework.

### 1.4 Scope and Contributions

This dissertation focuses on zero-shot EEG-to-image retrieval from natural-object responses in THINGS-EEG2 [4], considering intra-subject, cross-subject, and reduced-data evaluation. We introduce **Per-**



**Fig. 1.** Diagnostic-to-design overview. Three probes characterize recoverable visual content, sensor–time support, and visual-depth correspondence; their findings inform MVT and WSRA, which are evaluated through retrieval and representation analyses.

**ceptEEG**, a perceptual alignment framework with two complementary components. The *Mid-level Visual Target* (MVT) constructs compact visual supervision from an intermediate CLIP representation, reducing reliance on final-layer semantics. The *Windowed Spatiotemporal Residual Adapter* (WSRA) applies a bounded correction to the ATM representation using one diagnostic-motivated posterior early view and two broader contextual views.

On 200-way retrieval, PerceptEEG achieves  $65.85 \pm 7.46\%$  intra-subject and  $12.85 \pm 4.76\%$  cross-subject Recall@1. It obtains the highest intra-subject mean among the methods included in our comparison, whereas cross-subject performance remains below NeuroBridge. A staged ablation attributes the largest measured improvement to MVT and shows an additional WSRA gain when MVT is enabled. PerceptEEG also maintains a substantial fraction of its full-data performance under reduced training-concept coverage. The main contributions are summarized as follows:

- We develop a **controlled diagnostic protocol** that characterizes EEG–image representation mismatch across recoverable visual information, sensor–time support, and visual-encoder depth.
- We propose the **PerceptEEG framework**, which contains MVT to construct a compact perceptual supervision target and WSRA to refine EEG features through complementary sensor–time views within a lightweight residual architecture.
- We evaluate PerceptEEG on THINGS-EEG2, obtaining the highest intra-subject mean among the

included baselines at 65.85% **Recall@1**, together with 12.85% **cross-subject Recall@1** and improved performance under reduced training-concept coverage.

## 1.5 Dissertation Structure

The overall structure of this dissertation is illustrated in Fig. 1. Section 2 delineates the three diagnostic questions and details the evaluation procedures and principal findings. Section 3 presents the PerceptEEG framework, including WSRA, MVT, and the alignment objective. Section 4 specifies the experimental design and reports results for retrieval, reduced-data, ablation, and representation analyses. Section 5 examines the implications and limitations of the findings, and the final section synthesizes the conclusions and outlines directions for future research.

## 2 Diagnostic Study

### 2.1 Motivation and Diagnostic Questions

The research gap identified in Section 1.3 cannot be fully assessed from end-to-end retrieval performance alone. We therefore conduct a controlled diagnostic analysis using linear probing and systematic input restriction to characterize visual information recoverability under matched evaluation protocols. Specifically, we investigate three research questions:

- RQ1.** What visual information is linearly recoverable from pretrained EEG representations, and how does its recoverability compare with that of matched image representations?
- RQ2.** Which temporal intervals and channel groups support the recovery of visual information from EEG signals?
- RQ3.** Which level of the CLIP visual hierarchy exhibits the strongest correspondence with temporally restricted EEG representations?

Collectively, these questions characterize representational mismatch, the spatiotemporal sensitivity of EEG-based visual information, and the correspondence between EEG representations and different levels of the visual hierarchy. The resulting evidence informs the design choices in Section 3.

## 2.2 Diagnostic Method

### 2.2.1 RQ1: Visual Information Recoverability

**EEG representation extraction.** We evaluate six EEG encoders spanning different architectural families: ATM [10], DeepNet [16], EEGProject, EEGNet [17], ShallowNet [16], and TSConv [9]. All encoder parameters remain frozen during probing. For stimulus  $i$ , subject  $s$ , and repetition, we first extract an EEG embedding and then aggregate repeated trials at the stimulus level:

$$\mathbf{z}_{s,i}^E = \text{Norm} \left( \frac{1}{R_{s,i}} \sum_{r=1}^{R_{s,i}} \text{Norm}(f_{\text{enc}}(\mathbf{e}_{s,i,r})) \right), \quad (1)$$

where  $f_{\text{enc}}$  is the corresponding pretrained EEG encoder,  $\mathbf{e}_{s,i,r}$  is the EEG response, and  $R_{s,i}$  is the number of available repetitions. Stimulus-level aggregation prevents repeated presentations of the same image from being treated as independent examples and reduces trial-specific noise.

**Visual information targets.** We decompose visual information into three perceptual factors and one semantic factor. For image  $\mathbf{x}_i$ , the target for dimension  $m$  is

$$\mathbf{y}_i^m = g_m(\mathbf{x}_i), \quad m \in \{\text{color, structure, texture, semantics}\}. \quad (2)$$

Color is represented by marginal HSV histograms [26], structure by a histogram-of-oriented-gradients descriptor [27], and texture by concatenated local-binary-pattern [28], intensity-histogram, and gradient-magnitude descriptors. The THINGS [29] category label serves as the semantic target.

**Linear probing.** For each continuous target, we fit a ridge-regression probe [30]

$$\hat{\mathbf{y}}_{s,i}^m = \mathbf{W}_m \mathbf{z}_{s,i}^E + \mathbf{b}_m, \quad (3)$$

whereas semantic categories are predicted using a ridge classifier. Inputs and continuous targets are standardized using statistics computed exclusively from the training split. We compute variance-weighted  $R^2$ , mean dimension-wise Pearson correlation, and feature-level Recall@1 and Recall@5 for continuous targets, and Top-1 and Top-5 accuracy for semantics. The main comparisons in Table 1 and Fig. 2 use mean dimension-wise Pearson correlation for color, structure, and texture, and Top-1 accuracy for semantics.

**Evaluation protocols.** We distinguish subject-specific recoverability from cross-subject transfer. In the intra-subject protocol, a probe is trained and evaluated on image-disjoint splits from the same subject. In the cross-subject protocol, the probe is trained on all source subjects and evaluated on a held-out subject  $s$ :

$$\mathcal{D}_{\text{train}}^{(-s)} = \bigcup_{s' \neq s} \mathcal{D}_{s'}, \quad \mathcal{D}_{\text{test}}^{(s)} = \mathcal{D}_s. \quad (4)$$

The multi-encoder comparison in Table 1 uses Sub-01 as the target subject under both protocols. The detailed ATM analysis in Fig. 2 reports all ten subjects.

**Matched image reference.** To contextualize EEG recoverability, we apply an identical probing protocol to frozen OpenCLIP ViT-H/14 image embeddings, using the same stimuli, splits, targets, normalization, and probe settings. The primary ATM–image comparison uses 1024-dimensional representations for both modalities. Let  $S_{s,m,k}^E$  and  $S_{m,k}^I$  denote the EEG- and image-probe scores for subject  $s$ , visual dimension  $m$ , and metric  $k$ . We define the chance-adjusted relative recoverability ratio as

$$R_{\text{rel}}^{s,m,k} = \frac{S_{s,m,k}^E - S_{m,k}^{\text{floor}}}{S_{m,k}^I - S_{m,k}^{\text{floor}}}, \quad (5)$$

where  $S_{m,k}^{\text{floor}}$  is the corresponding chance or null level. Thus,  $R_{\text{rel}} = 1$  indicates equal probe performance after chance correction, whereas  $R_{\text{rel}} < 1$  indicates lower linear recoverability from EEG than from the matched image representation. For the primary comparison, the null Pearson correlation is 0, and semantic Top-1 chance is  $1/200 = 0.005$ . This ratio operationalizes relative probe-accessible information and is not an information-theoretic capacity measure.

### 2.2.2 RQ2: Spatiotemporal Recoverability

**Controlled sensor–time restriction.** To identify which temporal intervals and sensor groups support visual information recovery, we perform an input-restriction analysis using frozen ATM. For temporal window  $w$  and channel group  $c$ , let  $\mathbf{M}_{w,c} \in \{0, 1\}^{63 \times 250}$  be one within the selected sensor–time region and zero elsewhere. We construct

$$\tilde{\mathbf{e}}_{s,i}^{w,c} = \mathbf{e}_{s,i} \odot \mathbf{M}_{w,c}, \quad \mathbf{z}_{s,i}^{E,w,c} = f_{\text{ATM}}(\tilde{\mathbf{e}}_{s,i}^{w,c}). \quad (6)$$

Only the EEG input changes across conditions; ATM, the visual targets, the splits, and the probe settings remain fixed. Consequently, the comparison measures sensitivity to retained sensor–time inputs and should not be interpreted as neural source localization.

**Temporal and spatial conditions.** We evaluate four non-overlapping windows, 0–100, 100–300, 300–600, and 600–1000 ms, to identify intervals that independently support decoding. We additionally evaluate cumulative windows of 0–300, 0–600, and 0–1000 ms to characterize temporal integration. For the spatial analysis, we define frontal (15 channels), central (7), temporal (10), parietal (9), occipital (8), and posterior (24) channel groups, together with the complete 63-channel montage. The posterior group combines centroparietal, parietal, parieto-occipital, and occipital electrodes. We report the cross-subject LOSO results as the mean and standard deviation across the ten held-out subjects, using Pearson correlation for the three perceptual targets and Top-1 accuracy for semantics.

### 2.2.3 RQ3: Correspondence with the CLIP Visual Hierarchy

To examine which visual depth is most compatible with EEG, we sample Blocks 6, 14, 22, and 32 from a frozen OpenCLIP ViT-H/14 encoder. For stimulus  $i$ , the projected class token from block  $\ell$  defines the visual target

$$\mathbf{y}_i^{(\ell)} = g_{\text{CLIP}}^{(\ell)}(\mathbf{x}_i), \quad \ell \in \{6, 14, 22, 32\}. \quad (7)$$

For a temporal condition  $w$ , its EEG samples are average-pooled into  $B = 16$  temporal bins per channel and flattened to a fixed-dimensional predictor

$$\mathbf{q}_i^w = \text{vec}[\text{BinPool}_B(\mathbf{e}_i[:, \mathcal{T}_w])]. \quad (8)$$

A ridge regressor with regularization  $\lambda = 100$  maps  $\mathbf{q}_i^w$  to each CLIP target. Predictor and target statistics are estimated only from the training set; the predicted and target embeddings are then  $\ell_2$ -normalized for cosine-similarity retrieval. We use the Sub-01 intra-subject split and evaluate Recall@1 over a fixed 200-image gallery. The analysis covers the four independent windows from RQ2 and the three additional cumulative conditions 0–300, 0–600, and 0–1000 ms.

## 2.3 Diagnostic Results

### 2.3.1 RQ1: Visual Information Recoverability

**Recoverability across EEG encoders.** Table 1 shows that all six encoders achieve positive perceptual correlations and semantic Top-1 accuracy above the 0.5% chance level for the Sub-01 target. In the intra-subject setting, EEGNet gives the highest perceptual scores, whereas EEGProject gives the highest semantic accuracy. Under cross-subject transfer to Sub-01, ATM performs best on all four dimensions, reaching Pearson correlations of 0.171, 0.111, and 0.269 for color, structure, and texture, respectively, and 8.00% semantic Top-1 accuracy. We therefore select ATM in this preliminary

**Table 1.** Linear-probe performance across EEG encoders for Sub-01. Perceptual columns report mean dimension-wise Pearson correlation; Semantic reports Top-1 accuracy (%). Bold indicates the best result within each protocol.

| Intra-subject (Sub-01) |              |              |              |              | Cross-subject (held-out Sub-01) |              |              |              |             |
|------------------------|--------------|--------------|--------------|--------------|---------------------------------|--------------|--------------|--------------|-------------|
| Encoder                | Color        | Structure    | Texture      | Semantic     | Encoder                         | Color        | Structure    | Texture      | Semantic    |
| ATM                    | 0.195        | 0.116        | 0.283        | 30.80        | <b>ATM</b>                      | <b>0.171</b> | <b>0.111</b> | <b>0.269</b> | <b>8.00</b> |
| DeepNet                | 0.160        | 0.109        | 0.275        | 9.85         | DeepNet                         | 0.135        | 0.078        | 0.226        | 4.15        |
| <b>EEGProject</b>      | 0.186        | 0.110        | 0.252        | <b>52.40</b> | EEGProject                      | 0.076        | 0.056        | 0.156        | 1.75        |
| <b>EEGNet</b>          | <b>0.203</b> | <b>0.122</b> | <b>0.306</b> | 12.25        | EEGNet                          | 0.151        | 0.075        | 0.238        | 3.70        |
| ShallowNet             | 0.156        | 0.120        | 0.261        | 18.65        | ShallowNet                      | 0.119        | 0.080        | 0.210        | 4.45        |
| TSCnv                  | 0.143        | 0.104        | 0.238        | 18.05        | TSCnv                           | 0.111        | 0.075        | 0.210        | 3.40        |

Sub-01 comparison and keep it fixed for the subsequent ten-subject diagnostics. Because selection uses one target subject rather than the mean over all LOSO folds, it does not establish that ATM is universally optimal.

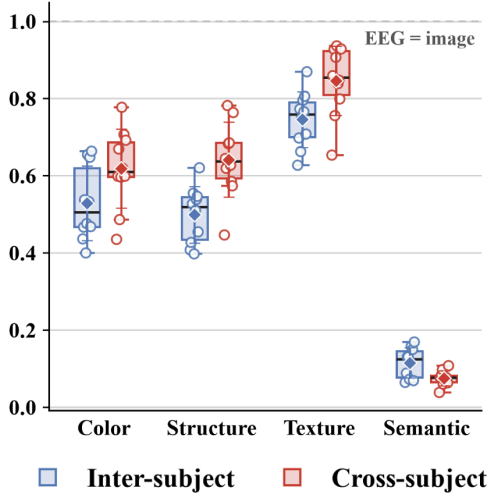
**EEG–image recoverability mismatch.** Figure 2 compares the chance-adjusted relative recoverability ratios for ATM over all ten subjects. All mean ratios remain below 1, indicating lower linear recoverability from EEG than from the matched image embedding. The cross-subject ratios for color, structure, and texture are  $0.618 \pm 0.102$ ,  $0.642 \pm 0.097$ , and  $0.846 \pm 0.090$  (mean  $\pm$  standard deviation), respectively, with texture closest to the image reference. Semantic matching is markedly weaker: the ratio is  $0.115 \pm 0.039$  under intra-subject evaluation and  $0.075 \pm 0.019$  under cross-subject evaluation, a relative reduction of 34.8%. The perceptual ratios do not show the same cross-subject decrease, suggesting that the measured transfer-related reduction is concentrated in the semantic probe under this diagnostic. The subject-level spread further indicates non-negligible inter-individual variation.

### 2.3.2 RQ2: Spatiotemporal Recoverability

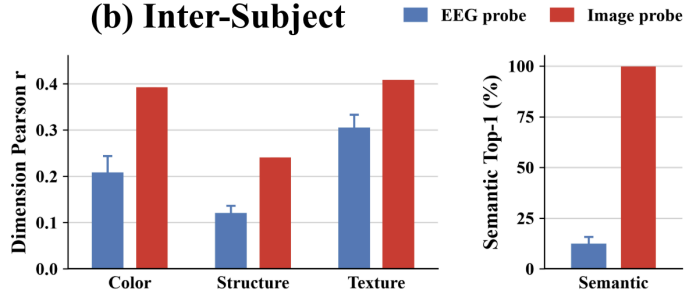
**Which temporal intervals support recovery?** Table 2 shows a strongly non-uniform temporal profile. Among the four non-overlapping windows, 100–300 ms yields the highest recoverability for every visual dimension, reaching 0.1881 for color, 0.1083 for structure, 0.2825 for texture, and 0.0400 for semantics. Performance falls sharply in the 300–600 ms interval and approaches the null or chance level during 600–1000 ms. These results identify 100–300 ms as the most informative independent interval under the present masking and probing protocol, while not excluding contributions from later activity in the unmasked signal.

**Which channel groups support recovery?** Restricting the 100–300 ms interval to posterior channels preserves most of the all-channel probe performance (Table 3). The posterior group reaches

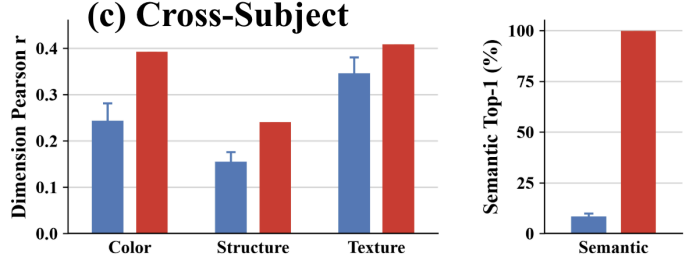
(a) EEG - Image Capacity Ratio



(b) Inter-Subject



(c) Cross-Subject



**Fig. 2.** ATM EEG-to-CLIP recoverability across ten subjects. (a) Chance-adjusted ratios using Pearson correlation for perceptual factors and Top-1 accuracy for semantics; the dashed line denotes equal recoverability. Circles are subjects, diamonds means, boxes interquartile ranges, and error bars one standard deviation. (b,c) Absolute probe performance.

**Table 2.** Cross-subject temporal-window probes using all 63 channels. Values are mean  $\pm$  standard deviation over ten LOSO folds. Perceptual columns report mean dimension-wise Pearson correlation; Semantic reports Top-1 accuracy. Bold marks the best non-overlapping window.

| Time window | Color                                 | Structure                             | Texture                               | Semantic                              |
|-------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| 0–100 ms    | 0.0604 $\pm$ 0.0106                   | 0.0520 $\pm$ 0.0116                   | 0.1482 $\pm$ 0.0230                   | 0.0084 $\pm$ 0.0023                   |
| 100–300 ms  | <b>0.1881 <math>\pm</math> 0.0285</b> | <b>0.1083 <math>\pm</math> 0.0141</b> | <b>0.2825 <math>\pm</math> 0.0218</b> | <b>0.0400 <math>\pm</math> 0.0085</b> |
| 300–600 ms  | 0.0283 $\pm$ 0.0097                   | 0.0301 $\pm$ 0.0078                   | 0.0686 $\pm$ 0.0189                   | 0.0105 $\pm$ 0.0025                   |
| 600–1000 ms | 0.0016 $\pm$ 0.0070                   | 0.0023 $\pm$ 0.0031                   | 0.0065 $\pm$ 0.0066                   | 0.0041 $\pm$ 0.0013                   |

0.1839, 0.1033, 0.2781, and 0.0344 for color, structure, texture, and semantics, corresponding to 97.7%, 95.4%, 98.4%, and 86.0% of the respective all-channel scores. Occipital channels alone also retain substantial perceptual recoverability, whereas frontal and central restrictions yield scores near chance level. Thus, the diagnostic supports posterior-dominant input sensitivity during the peak interval. Because the channel groups differ in size and scalp EEG is subject to volume conduction, these comparisons do not establish additive regional contributions or provide anatomical source localization.

### 2.3.3 RQ3: Correspondence with the CLIP Visual Hierarchy

Table 4 shows that 100–300 ms provides the strongest independent-window correspondence for every sampled CLIP block. Within this interval, Block 14 achieves the highest Recall@1 of 0.365, compared with 0.275, 0.160, and 0.120 for Blocks 6, 22, and 32, respectively. Block 14 is also best for all three cumulative windows, reaching 0.325, 0.305, and 0.240 for 0–300, 0–600, and 0–1000 ms. Overall, it ranks first in six of the seven reported temporal conditions; only the late 600–1000 ms window favors Block 22, and all methods are near chance in that interval. These results indicate that temporally restricted

**Table 3.** Cross-subject channel-group probes at 100–300 ms, averaged over ten LOSO folds. Perceptual columns report mean dimension-wise Pearson correlation; Semantic reports Top-1 accuracy. Bold marks the best restricted group; *All* is the 63-channel reference and Posterior is an overlapping composite.

| Channel group | #Ch. | Color         | Structure     | Texture       | Semantic      |
|---------------|------|---------------|---------------|---------------|---------------|
| Frontal       | 15   | 0.0065        | 0.0011        | 0.0092        | 0.0048        |
| Central       | 7    | 0.0050        | 0.0002        | 0.0062        | 0.0052        |
| Temporal      | 10   | 0.0125        | 0.0147        | 0.0236        | 0.0090        |
| Parietal      | 9    | 0.0319        | 0.0391        | 0.0799        | 0.0131        |
| Occipital     | 8    | 0.1786        | 0.0944        | 0.2687        | 0.0288        |
| Posterior     | 24   | <b>0.1839</b> | <b>0.1033</b> | <b>0.2781</b> | <b>0.0344</b> |
| <i>All</i>    | 63   | 0.1881        | 0.1083        | 0.2825        | 0.0400        |

**Table 4.** Sub-01 intra-subject Recall@1 across sampled CLIP blocks and EEG windows. EEG is pooled into 16 temporal bins and mapped to each projected CLIP class token by ridge regression. The gallery contains 200 images (chance = 0.005); bold marks the best block per window.

| Block    | 0–100        | Independent  |              |              | Cumulative   |              |              |
|----------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|          |              | 100–300      | 300–600      | 600–1000     | 0–300        | 0–600        | 0–1000       |
| Block 6  | 0.080        | 0.275        | 0.070        | 0.005        | 0.260        | 0.195        | 0.130        |
| Block 14 | <b>0.085</b> | <b>0.365</b> | <b>0.095</b> | 0.005        | <b>0.325</b> | <b>0.305</b> | <b>0.240</b> |
| Block 22 | 0.035        | 0.160        | 0.090        | <b>0.015</b> | 0.150        | 0.175        | 0.105        |
| Block 32 | 0.045        | 0.120        | 0.045        | 0.000        | 0.130        | 0.105        | 0.090        |

EEG is more compatible with the sampled intermediate CLIP representation than with the final block. This pattern is consistent with the smaller perceptual than semantic recoverability mismatch observed in RQ1, but does not imply that a CLIP block corresponds to a single type of visual information.

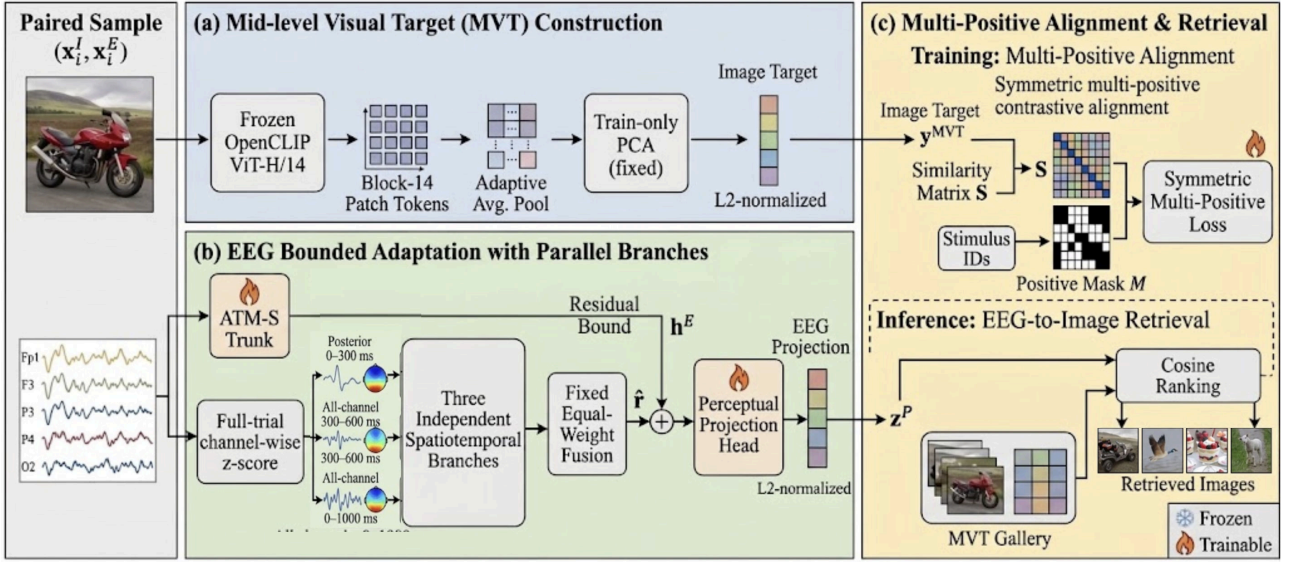
### 2.3.4 Summary

RQ1 indicates a pronounced semantic EEG–image recoverability mismatch, RQ2 suggests peak input sensitivity in 100–300 ms with a posterior emphasis, and RQ3 observes the strongest sampled visual correspondence at the mid-level CLIP representation. Together, these task-specific findings inform the visual-depth and sensor–time priors developed next.

## 3 PerceptEEG Framework

### 3.1 Motivation and Problem Formulation

Building on the diagnostic findings in Section 2, PerceptEEG translates the observed representation and spatiotemporal characteristics into explicit design priors. RQ1 and RQ3 suggest that an intermediate visual representation provides a more suitable target than relying exclusively on the final CLIP embedding, motivating the use of an intermediate visual target in MVT. MVT further specifies



**Fig. 3.** PerceptEEG overview. WSRA applies bounded sensor–time residuals to the ATM feature, while MVT pools and PCA-compresses Block 14 OpenCLIP patch tokens using a training-fitted transform. A projection head aligns EEG and MVT embeddings through a symmetric multi-positive contrastive loss.

how patch tokens at this depth are pooled and compressed for cross-modal alignment. RQ2 motivates the early posterior view adopted in WSRA, while its additional temporal views are included to retain complementary information beyond the most informative interval. As illustrated in Fig. 3, these components are jointly optimized through a stimulus-aware multi-positive contrastive objective. This objective is introduced as an independent training design to enforce discriminative EEG–image alignment, rather than being a direct consequence of any single diagnostic finding.

Let

$$\mathcal{D} = \left\{ \left( x_i^E, x_i^I, s_i, q_i \right) \right\}_{i=1}^N \quad (9)$$

denote the paired data, where  $x_i^E \in \mathbb{R}^{C \times T}$  is an EEG observation,  $x_i^I$  is its stimulus image,  $s_i$  is the subject identity, and  $q_i \in \mathcal{Q}$  is the exact stimulus identity. We use  $C = 63$  channels and  $T = 250$  samples acquired at 250 Hz. Let  $\mathcal{C}_{\text{train}}$  and  $\mathcal{C}_{\text{test}}$  denote the concept sets associated with the training and test stimuli, respectively. The two concept sets are disjoint,

$$\mathcal{C}_{\text{train}} \cap \mathcal{C}_{\text{test}} = \emptyset. \quad (10)$$

The test images constitute the retrieval gallery. Given an EEG query, the task is to rank gallery images according to the similarity between its predicted perceptual embedding and their precomputed MVT representations.

## 3.2 EEG Representation Learning

### 3.2.1 EEG Backbone

In the preliminary encoder comparison in Section 2.3, ATM exhibits more consistent performance across visual dimensions and evaluation settings. We therefore adopt ATM as the EEG backbone. Given an EEG trial  $x_i^E \in \mathbb{R}^{63 \times 250}$ , ATM first maps the input to a token sequence and incorporates subject information through a subject token. A single-layer iTransformer [31] then performs channel-wise attention over the EEG tokens. After attention, the subject token is removed and all 63 EEG tokens are passed to the temporal–spatial convolutional module, which applies temporal convolution, average pooling, spatial convolution, and a  $1 \times 1$  projection. We write the ATM forward pass as

$$u_i^E = f_{\text{attn}} \left( f_{\text{emb}}(x_i^E, s_i) \right), \quad (11)$$

$$h_i^E = f_{\text{flat}} \left( f_{\text{conv}}(u_i^E) \right) \in \mathbb{R}^{1440}, \quad (12)$$

and

$$z_i^E = f_{\text{proj}}(h_i^E) \in \mathbb{R}^{1024}, \quad (13)$$

where  $f_{\text{emb}}$ ,  $f_{\text{attn}}$ ,  $f_{\text{conv}}$ ,  $f_{\text{flat}}$ , and  $f_{\text{proj}}$  denote the embedding, channel-wise attention, temporal–spatial convolution, flattening, and original MLP projection operations, respectively. PerceptEEG is initialized from the pretrained ATM and preserves its backbone architecture. The 1440-dimensional feature  $h_i^E$  provides the feature space for WSRA, whereas the original readout  $z_i^E$  is retained only as the reference ATM output and is not used in the PerceptEEG retrieval score. During training, ATM parameters up to  $h_i^E$  are optimized jointly with the newly introduced modules. For intra-subject training, the model uses the corresponding subject token. For cross-subject evaluation, all source-subject training trials and held-out-subject test trials use the same shared token, preventing the model from accessing held-out subject identity information through the subject token.

### 3.2.2 Windowed Spatiotemporal Residual Adapter

The ATM backbone aggregates the complete EEG trial without explicitly separating sensor–time views. RQ2 shows that the 100–300 ms interval is the most informative independent window and that posterior channels retain 86.0–98.4% of the corresponding all-channel probe performance across the four visual dimensions. Motivated by this input-sensitivity pattern, WSRA introduces an early posterior view and combines it with two broader views for complementary coverage. These ranges are architectural priors rather than anatomical source-localization claims.

**Full-trial channel normalization and sensor–time views.** The ATM trunk consumes the original preprocessed EEG, whereas the adapter standardizes its input channel-wise over the complete trial,

$$\begin{aligned}\mu_{i,c} &= \frac{1}{T} \sum_{t=1}^T x_{i,c,t}^E, \\ \sigma_{i,c} &= \sqrt{\frac{1}{T} \sum_{t=1}^T (x_{i,c,t}^E - \mu_{i,c})^2}, \\ \bar{x}_{i,c,t}^E &= \frac{x_{i,c,t}^E - \mu_{i,c}}{\max(\sigma_{i,c}, \epsilon)}.\end{aligned}\tag{14}$$

Here,  $\mu_{i,c}$  and  $\sigma_{i,c}$  are the temporal mean and standard deviation of channel  $c$  within trial  $i$ , and  $\epsilon = 10^{-5}$  ensures numerical stability. We then slice three views from  $\bar{x}_i^E$ :

$$\begin{aligned}v_i^{(1)} &= \bar{x}_i^E[\mathcal{E}_{\text{post}}, 0:75], & \text{posterior, 0–300 ms,} \\ v_i^{(2)} &= \bar{x}_i^E[:, 75:150], & \text{all channels, 300–600 ms,} \\ v_i^{(3)} &= \bar{x}_i^E[:, 0:250], & \text{all channels, 0–1000 ms.}\end{aligned}\tag{15}$$

The posterior set  $\mathcal{E}_{\text{post}}$  contains the 24 electrodes assigned to the centroparietal, parietal, parieto-occipital, and occipital groups (the CP, P, PO, and O prefixes). The first view extends the diagnostic peak interval to 0–300 ms to preserve the preceding onset response. The second and third views provide later and full-trial context, reducing reliance on a single restricted view.

**Residual branch.** Each view is processed by an independent temporal–spatial branch  $g_k$ . Given  $v_i^{(k)}$ , the branch applies a depth-wise temporal convolution with kernel size 9 and GELU [32], a  $1 \times 1$  convolution that mixes sensors into 32 hidden channels, and a depth-wise context convolution with kernel size 5 whose activated output is added residually. Adaptive average pooling reduces the temporal axis to four samples. The resulting  $32 \times 4$  feature is flattened, layer-normalized [33], projected to rank 64, and mapped to the 1440-dimensional ATM feature space:

$$\begin{aligned}a_i^{(k)} &= \text{GELU}\left(\text{DWConv}_9^{(k)}\left(v_i^{(k)}\right)\right), \\ b_i^{(k)} &= \text{GELU}\left(\text{PWConv}_1^{(k)}\left(a_i^{(k)}\right)\right), \\ c_i^{(k)} &= b_i^{(k)} + \text{GELU}\left(\text{DWConv}_5^{(k)}\left(b_i^{(k)}\right)\right), \\ r_i^{(k)} &= W_{\text{up}}^{(k)} \text{GELU}\left[W_{\text{down}}^{(k)} \text{LN}\left(\text{vec}\left(\text{AAP}_4\left(c_i^{(k)}\right)\right)\right) + b_{\text{down}}^{(k)}\right] + b_{\text{up}}^{(k)} \in \mathbb{R}^{1440}.\end{aligned}\tag{16}$$

In the reported configuration,  $W_{\text{down}}^{(k)} \in \mathbb{R}^{64 \times 128}$  and  $W_{\text{up}}^{(k)} \in \mathbb{R}^{1440 \times 64}$ , and branch dropout is disabled. Each  $W_{\text{up}}^{(k)}$  and its bias are initialized to zero, so the refined feature is identical to the ATM feature at initialization.

**Fixed fusion and sample-wise residual bound.** The three branch outputs are combined using fixed, identical residual-scaling coefficients,

$$r_i^{\text{raw}} = \sum_{k=1}^3 \alpha_k r_i^{(k)}, \quad \alpha_1 = \alpha_2 = \alpha_3 = 0.1. \quad (17)$$

The coefficients are not normalized mixture weights and are therefore not constrained to sum to one. Instead, the magnitude of the fused correction is separately bounded relative to the pre-adaptor ATM feature:

$$\begin{aligned} n_i^h &= \max(\|h_i^E\|_2, \epsilon), \\ n_i^r &= \max(\|r_i^{\text{raw}}\|_2, \epsilon), \\ \eta_i &= \min\left(1, \rho \frac{n_i^h}{n_i^r}\right), \quad \rho = 0.3, \\ \hat{r}_i &= \eta_i r_i^{\text{raw}}, \quad \tilde{h}_i^E = h_i^E + \hat{r}_i. \end{aligned} \quad (18)$$

Up to the numerical clamp, the additive correction is therefore at most 30% of the base-feature norm. Together with zero initialization, this bounded formulation starts from the ATM pathway and learns a controlled feature-level adaptation.

### 3.3 Mid-level Visual Target

The visual alignment depth is chosen to reflect the diagnostic evidence rather than defaulting to the final CLIP representation. RQ1 shows that the semantic EEG–image recoverability mismatch is substantially larger than the perceptual mismatch, and the Sub-01 analysis in RQ3 finds that Block 14 yields the strongest EEG correspondence among the sampled blocks in six of the seven tested temporal conditions. We therefore select Block 14 as the visual depth for MVT and use a frozen OpenCLIP ViT-H/14 [34] encoder with the `laion2b_s32b_b79k` checkpoint [35]. Constructing MVT from patch tokens, applying  $2 \times 2$  pooling, and fitting PCA are separate target-design choices that preserve a compact spatially pooled representation and are evaluated through the downstream ablation rather than directly by RQ3. Let  $U_i^{(14)}$  denote the token sequence output by Block 14. We remove the class token, restore the patch tokens to a square grid, and pool the grid to  $2 \times 2$ :

$$p_i^{\text{I}} = \text{vec}\left(\text{AAP}_{2 \times 2}\left[\text{Grid}\left(U_{i,\text{patch}}^{(14)}\right)\right]\right) \in \mathbb{R}^{4D_v}, \quad (19)$$

where  $D_v$  is the visual-transformer width. This produces a compact feature that preserves a coarse  $2 \times 2$  spatial layout before dimensionality reduction. PCA is fitted only on the pooled features of the unique images in the training split. Let  $\mu_{\text{PCA}}$  and  $W_{\text{PCA}} \in \mathbb{R}^{d_p \times 4D_v}$  denote the fitted mean and

projection matrix. The MVT representation is

$$y_i^{\text{MVT}} = \frac{W_{\text{PCA}} (p_i^{\text{I}} - \mu_{\text{PCA}})}{\|W_{\text{PCA}} (p_i^{\text{I}} - \mu_{\text{PCA}})\|_2} \in \mathbb{S}^{d_p-1}. \quad (20)$$

We set  $d_p = \min(256, n_{\text{train}}^{\text{img}} - 1, 4D_v)$ , which gives  $d_p = 256$  in the standard full-data setting. The OpenCLIP encoder and fitted PCA transform are fixed thereafter; no validation or test image is used to fit PCA. We use randomized SVD [36] with random seed 0 and do not whiten the PCA coordinates.

### 3.4 Perceptual Alignment

After WSRA refines the ATM feature and MVT constructs the image target, we map both modalities into the same unit-normalized perceptual space and optimize their correspondence with a symmetric contrastive objective.

#### 3.4.1 Perceptual EEG Projection

A dedicated projection head maps the WSRA-refined EEG feature to the MVT space. With  $d_h = 1440$  and hidden width 512, it computes

$$\begin{aligned} e_i^{\text{P}} &= W_2 \text{Dropout}_{0.1} \left( \text{GELU} \left( W_1 \text{LN}(\tilde{h}_i^{\text{E}}) + b_1 \right) \right) + b_2, \\ z_i^{\text{P}} &= \frac{e_i^{\text{P}}}{\|e_i^{\text{P}}\|_2} \in \mathbb{S}^{d_p-1}, \end{aligned} \quad (21)$$

where  $W_1 \in \mathbb{R}^{512 \times 1440}$  and  $W_2 \in \mathbb{R}^{d_p \times 512}$ . PerceptEEG uses  $z_i^{\text{P}}$  as its EEG representation for contrastive training and retrieval; the original ATM final embedding is not fused into the retrieval score.

#### 3.4.2 Symmetric Multi-positive Contrastive Loss

For a mini-batch of  $B$  EEG–image pairs, we form the score matrix

$$S_{jk} = \gamma \left( z_j^{\text{P}} \right)^{\top} y_k^{\text{MVT}}, \quad \gamma = \min(\exp(\ell), 100), \quad (22)$$

where the learnable log-scale is initialized as  $\ell = \log(1/0.07)$ . Because multiple EEG observations can correspond to the same image, a diagonal-only target would treat valid matches as negatives. We therefore define the in-batch positive sets

$$\mathcal{P}_j^{E \rightarrow I} = \{k \mid q_k = q_j\}, \quad \mathcal{P}_k^{I \rightarrow E} = \{j \mid q_j = q_k\}. \quad (23)$$

The two retrieval directions are optimized as

$$\mathcal{L}_{E \rightarrow I} = -\frac{1}{B} \sum_{j=1}^B \log \frac{\sum_{k \in \mathcal{P}_j^{E \rightarrow I}} \exp(S_{jk})}{\sum_{k=1}^B \exp(S_{jk})}, \quad (24)$$

and

$$\mathcal{L}_{I \rightarrow E} = -\frac{1}{B} \sum_{k=1}^B \log \frac{\sum_{j \in \mathcal{P}_k^{I \rightarrow E}} \exp(S_{jk})}{\sum_{j=1}^B \exp(S_{jk})}. \quad (25)$$

The perceptual alignment objective is

$$\mathcal{L}_{\text{perc}} = \frac{1}{2} (\mathcal{L}_{E \rightarrow I} + \mathcal{L}_{I \rightarrow E}). \quad (26)$$

The positive sets are batch-local: additional observations act as positives only when trials with the same exact stimulus occur in the same mini-batch. If every stimulus occurs once, Eq. ((26)) reduces to the standard symmetric one-positive contrastive loss.

### 3.4.3 Residual Regularization and Total Objective

The architectural cap bounds the largest fused correction, while two magnitude penalties encourage economical residuals during optimization. We penalize the post-cap residual and the unweighted branch outputs as

$$\mathcal{L}_{\text{res}} = \frac{1}{Bd_h} \sum_{i=1}^B \|\hat{r}_i\|_2^2, \quad \mathcal{L}_{\text{branch}} = \frac{1}{3Bd_h} \sum_{i=1}^B \sum_{k=1}^3 \|r_i^{(k)}\|_2^2. \quad (27)$$

The second term discourages large pre-fusion branch outputs that could otherwise be attenuated by the fixed weights or cancel across branches.

We additionally regularize the residual across repeated presentations of the same image within the current mini-batch. Let

$$\mathcal{R}_{q,B} = \{i \in \{1, \dots, B\} \mid q_i = q\}, \quad \mathcal{Q}_B = \{q \mid |\mathcal{R}_{q,B}| > 1\}, \quad (28)$$

and define

$$\bar{r}_{q,B} = \frac{1}{|\mathcal{R}_{q,B}|} \sum_{i \in \mathcal{R}_{q,B}} \hat{r}_i. \quad (29)$$

The repetition-consistency term is

$$\mathcal{L}_{\text{rep}} = \begin{cases} \frac{1}{|\mathcal{Q}_B|} \sum_{q \in \mathcal{Q}_B} \frac{1}{|\mathcal{R}_{q,B}| d_h} \sum_{i \in \mathcal{R}_{q,B}} \|\hat{r}_i - \bar{r}_{q,B}\|_2^2, & |\mathcal{Q}_B| > 0, \\ 0, & |\mathcal{Q}_B| = 0. \end{cases} \quad (30)$$

This prior acts only on the additive residual, rather than forcing the complete EEG representation to be invariant across trials or subjects.

The complete PerceptEEG objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{perc}} + 10^{-4} \mathcal{L}_{\text{res}} + 10^{-5} \mathcal{L}_{\text{branch}} + 10^{-4} \mathcal{L}_{\text{rep}}. \quad (31)$$

These coefficients assign the three regularizers an auxiliary role relative to perceptual alignment. The ATM layers up to  $h_i^E$ , WSRA branch parameters, perceptual projection head, and logit scale are optimized jointly. The three branch fusion weights are fixed. The original ATM readout, frozen OpenCLIP encoder, and training-fitted PCA transform receive no gradient from Eq. ((31)).

### 3.5 Implementation and Inference

The full-data configuration is trained for 40 epochs using AdamW [37] with a learning rate of  $3 \times 10^{-4}$  and weight decay of  $10^{-2}$ . Mini-batch sizes are 1024 for intra-subject training and 256 for cross-subject training. All four training repetitions per image are retained as individual training observations, and the implementation uses bfloat16 automatic mixed precision. These settings are held fixed between the intra-subject and cross-subject variants except for batch size and the subject-token policy described above. In the standard  $d_p = 256$  setting, WSRA adds 313,158 trainable parameters and the perceptual projection head adds 872,000.

At inference,  $y_j^{\text{MVT}}$  is precomputed for every gallery image using the frozen visual pipeline and the PCA transform fitted on the corresponding training split. Repeated raw EEG observations of each test stimulus are averaged before encoding, yielding one query per gallery stimulus under the retrieval evaluation protocol. We use signal-space averaging for retrieval to form one low-noise query per test stimulus, while retaining embedding-space aggregation in the diagnostic study to characterize frozen trial-level representations. For an EEG query  $x_i^E$ , PerceptEEG ranks candidate image  $j$  by

$$\text{score}(x_i^E, x_j^I) = (z_i^P)^\top y_j^{\text{MVT}}, \quad \hat{j} = \arg \max_j (z_i^P)^\top y_j^{\text{MVT}}. \quad (32)$$

Both vectors are unit-normalized, so the score is cosine similarity. The positive logit scale in Eq. ((22))

is omitted because a shared scalar does not affect the ranking. No semantic-score fusion, parameter update, or loss computation is used during inference.

## 4 Experiments and Results

### 4.1 Overview

We evaluate PerceptEEG from four complementary perspectives. We first compare 200-way zero-shot retrieval with representative EEG–image alignment methods and then test the practical low-data setting emphasized in the Introduction. A staged ablation measures the increment from MVT and the conditional effect of adding WSRA on top of MVT, after which a native-space analysis examines the learned cross-modal geometry. Qualitative examples and two exploratory extensions provide additional diagnostic context. Unless stated otherwise, results are aggregated over all ten subjects.

### 4.2 Implementation Details

#### 4.2.1 Datasets and Preprocessing

We evaluate on THINGS-EEG2 [4], which contains EEG recordings from ten subjects viewing natural-object images. The training split contains 1,654 concepts with ten images per concept and four EEG repetitions per image. The test split contains one image from each of 200 disjoint concepts, with 80 EEG repetitions per image. Of the 64 recorded channels, we retain 63 EEG channels as model input. Signals are band-pass filtered to 0.1–100 Hz, epoched over 0–1000 ms after stimulus onset, baseline-corrected using the 200 ms pre-stimulus interval, downsampled from 1000 to 250 Hz, and multivariate-noise-normalized [38] using training data. Each trial is therefore represented as a  $63 \times 250$  tensor. All methods use the same train/test concepts and the same preprocessing.

#### 4.2.2 Evaluation Protocol and Training Details

We consider two protocols. In the intra-subject setting, a subject-specific model is trained and evaluated on the same participant using disjoint training and test concepts. In the cross-subject setting, strict leave-one-subject-out (LOSO) evaluation trains on nine participants and tests on the unseen tenth participant; no held-out-subject EEG is used for training. Both protocols rank the same 200-image test gallery. As described in Section 3.5, the 80 repeated test trials are averaged in signal space before encoding.

**Table 5.** THINGS-EEG2 200-way Top-1 and Top-5 retrieval accuracy (%). Values are per-subject scores and mean  $\pm$  sample standard deviation over ten subjects. Bold and underlining indicate the best and second-best results, respectively, within each protocol and metric.

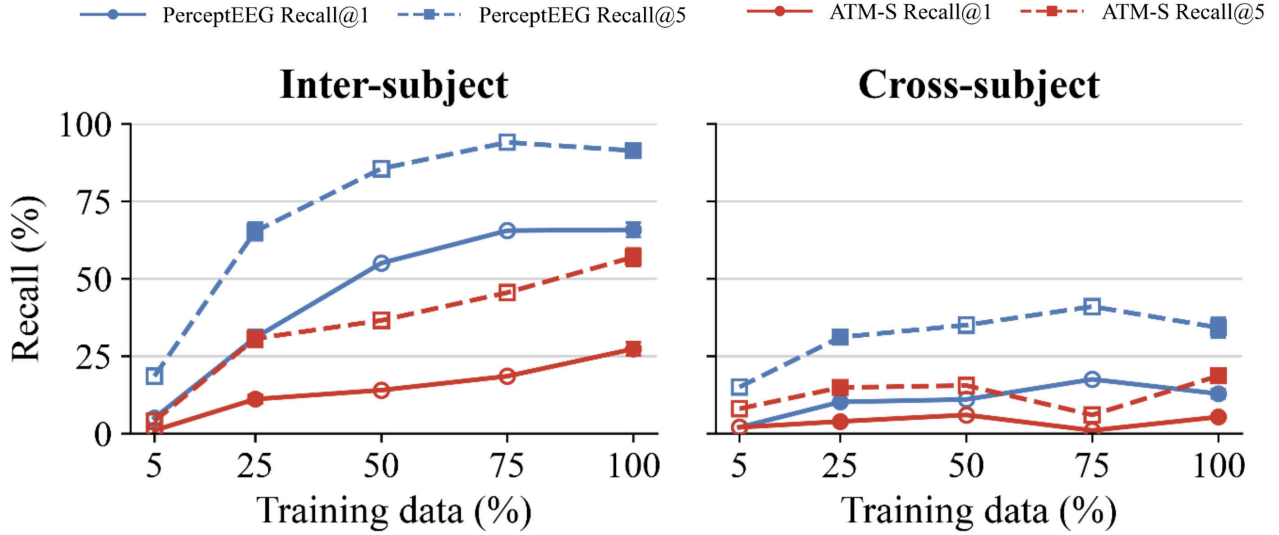
| Method                                               | Venue        | Metric | Sub-01      | Sub-02      | Sub-03      | Sub-04      | Sub-05      | Sub-06      | Sub-07      | Sub-08      | Sub-09      | Sub-10      | Mean $\pm$ Std.                    |
|------------------------------------------------------|--------------|--------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------------------------|
| Intra-Subject: subject-specific training and testing |              |        |             |             |             |             |             |             |             |             |             |             |                                    |
| NICE                                                 | ICLR 2024    | Top-1  | 13.2        | 13.5        | 14.5        | 20.6        | 10.1        | 16.5        | 17.0        | 22.9        | 15.4        | 17.4        | 16.11 $\pm$ 3.70                   |
|                                                      |              | Top-5  | 39.5        | 40.3        | 42.7        | 52.7        | 31.5        | 44.0        | 42.1        | 56.1        | 41.6        | 45.8        | 43.63 $\pm$ 6.87                   |
| ATM                                                  | NeurIPS 2024 | Top-1  | 25.6        | 22.0        | 25.0        | 31.4        | 12.9        | 21.3        | 30.5        | 38.8        | 34.4        | 29.1        | 27.10 $\pm$ 7.40                   |
|                                                      |              | Top-5  | 60.4        | 54.5        | 62.4        | 60.9        | 43.0        | 51.1        | 61.5        | 72.0        | 51.5        | 63.5        | 58.08 $\pm$ 8.16                   |
| UBP                                                  | CVPR 2025    | Top-1  | 41.2        | 51.2        | 51.2        | 51.1        | 42.2        | 57.5        | 49.0        | 58.6        | 45.1        | 61.5        | 50.86 $\pm$ 6.85                   |
|                                                      |              | Top-5  | 70.5        | 80.9        | 82.0        | 76.9        | 72.8        | 83.5        | 79.9        | 85.8        | 76.2        | 88.2        | 79.67 $\pm$ 5.61                   |
| NeuroBridge                                          | AAAI 2026    | Top-1  | 50.0        | 63.2        | 61.6        | 61.4        | <b>54.8</b> | <b>69.7</b> | 62.7        | 71.2        | <b>64.0</b> | <b>73.6</b> | <u>63.22 <math>\pm</math> 7.19</u> |
|                                                      |              | Top-5  | 77.6        | 90.6        | 91.1        | <b>90.0</b> | <b>85.0</b> | <b>92.9</b> | 88.8        | 95.1        | <b>91.0</b> | <b>97.1</b> | <u>89.92 <math>\pm</math> 5.45</u> |
| PerceptEEG                                           | Ours         | Top-1  | <b>77.0</b> | <b>65.5</b> | <b>64.0</b> | <b>64.5</b> | 50.5        | 61.5        | <b>70.0</b> | <b>74.5</b> | 62.0        | 69.0        | <b>65.85 <math>\pm</math> 7.46</b> |
|                                                      |              | Top-5  | <b>98.0</b> | <b>93.5</b> | <b>91.5</b> | 88.5        | 81.0        | 87.0        | <b>96.0</b> | <b>95.5</b> | 89.0        | 94.0        | <b>91.40 <math>\pm</math> 5.10</b> |
| Cross-Subject: leave-one-subject-out evaluation      |              |        |             |             |             |             |             |             |             |             |             |             |                                    |
| NICE                                                 | ICLR 2024    | Top-1  | 7.6         | 5.9         | 6.0         | 6.3         | 4.4         | 5.6         | 5.6         | 6.3         | 5.7         | 8.4         | 6.18 $\pm$ 1.11                    |
|                                                      |              | Top-5  | 22.8        | 20.5        | 22.3        | 20.7        | 18.3        | 22.2        | 19.7        | 22.0        | 17.6        | 28.3        | 21.44 $\pm$ 2.97                   |
| ATM                                                  | NeurIPS 2024 | Top-1  | 10.5        | 7.1         | 11.9        | 14.7        | 7.0         | 11.1        | <b>16.1</b> | 15.0        | 4.9         | 20.5        | 11.88 $\pm$ 4.80                   |
|                                                      |              | Top-5  | 26.8        | 24.8        | 33.8        | 39.4        | 23.9        | 35.8        | 43.5        | 40.3        | 22.7        | 46.5        | 33.75 $\pm$ 8.72                   |
| UBP                                                  | CVPR 2025    | Top-1  | 11.5        | 15.5        | 9.8         | 13.0        | 8.8         | 11.7        | 10.2        | 12.2        | <b>15.5</b> | 16.0        | 12.42 $\pm$ 2.55                   |
|                                                      |              | Top-5  | 29.7        | 40.0        | 27.0        | 32.3        | 33.8        | 31.0        | 23.8        | 32.2        | <b>40.5</b> | 43.5        | 33.38 $\pm$ 6.25                   |
| NeuroBridge                                          | AAAI 2026    | Top-1  | <b>23.2</b> | <b>21.2</b> | <b>13.2</b> | <b>17.0</b> | <b>14.5</b> | <b>25.0</b> | 15.3        | <b>20.1</b> | 13.7        | <b>27.2</b> | <b>19.04 <math>\pm</math> 5.02</b> |
|                                                      |              | Top-5  | <b>52.4</b> | <b>49.3</b> | <b>36.5</b> | <b>45.3</b> | <b>37.7</b> | <b>55.0</b> | <b>45.1</b> | <b>44.9</b> | 36.5        | <b>56.3</b> | <b>45.90 <math>\pm</math> 7.38</b> |
| PerceptEEG                                           | Ours         | Top-1  | 16.5        | 17.0        | 7.5         | 10.0        | 10.0        | 8.5         | 13.5        | 14.0        | 9.0         | 22.5        | <u>12.85 <math>\pm</math> 4.76</u> |
|                                                      |              | Top-5  | 36.5        | 40.5        | 26.0        | 23.5        | 33.5        | 26.0        | 42.5        | 30.0        | 28.0        | 55.5        | <u>34.20 <math>\pm</math> 9.85</u> |

Our implementation uses PyTorch [39] and is trained on NVIDIA A100-SXM4-80GB and L40S GPUs. All configurations run for 40 epochs with AdamW, an initial learning rate of  $3 \times 10^{-4}$ , and weight decay  $10^{-2}$ . The learnable logit scale is initialized with temperature  $\tau = 0.07$ . Batch sizes are 1,024 for intra-subject training and 256 for cross-subject training. We use bfloat16 automatic mixed precision and TF32 matrix multiplication. Full-data and LOSO results use the epoch-40 checkpoint. For the 25% intra-subject condition, checkpoint selection uses validation MRR on an image-disjoint validation split.

### 4.3 Retrieval Performance

Table 5 compares PerceptEEG with NICE [9], ATM [10], UB [14], and NeuroBridge [15] on 200-way zero-shot retrieval. We report Top-1 and Top-5 accuracy for each subject and their mean  $\pm$  standard deviation over the ten subjects. PerceptEEG substantially improves intra-subject retrieval but remains below the strongest cross-subject baseline.

**Intra-subject results.** PerceptEEG achieves the best reported mean among the compared methods, with  $65.85 \pm 7.46\%$  Top-1 and  $91.40 \pm 5.10\%$  Top-5 accuracy. It exceeds NeuroBridge, the strongest baseline in this setting, by 2.63 and 1.48 percentage points, respectively. Relative to ATM, the corresponding gains are 38.75 and 33.32 points. PerceptEEG attains the highest Top-1 accuracy



**Fig. 4.** Recall@1 and Recall@5 versus retained training-concept coverage for PerceptEEG and the reduced-data ATM baseline. The 200-image test gallery is fixed across all fractions.

for six subjects and the highest Top-5 accuracy for five, so the mean gain is not attributable to one participant. Because each model is trained with one seed, however, the result does not quantify optimization uncertainty.

**Cross-subject results.** Performance decreases for all methods under transfer to an unseen participant. PerceptEEG reaches  $12.85 \pm 4.76\%$  Top-1 and  $34.20 \pm 9.85\%$  Top-5, ranking second on both metrics. It is 0.43 points above UBP in Top-1 and 0.45 points above ATM in Top-5, but remains 6.19 and 11.70 points below NeuroBridge.

#### 4.4 Reduced-Data Evaluation

We reduce training diversity at the concept level rather than randomly discarding individual trials. For each retained concept, all ten image exemplars and all four EEG trials per exemplar are preserved, while the 200-image test gallery remains fixed. Table 6 compares the pre-specified 25% operating point with full-data training, and Fig. 4 additionally reports Recall@1 and Recall@5 over 5%, 25%, 50%, 75%, and 100% training-concept coverage. The ATM values in this section are controlled runs from the reduced-data sweep; the 100% endpoint is therefore distinct from the published ATM result in Table 5.

At 25% training coverage, PerceptEEG exceeds ATM by 19.85 and 34.60 percentage points in intra-subject Recall@1 and Recall@5, respectively. Its 65.25% Recall@5 also exceeds the 56.95% obtained by ATM with all training concepts. Under cross-subject LOSO, the corresponding 25% gains are 6.30 and 16.25 points. Relative to its own full-data result, PerceptEEG retains 79.4% of Recall@1

**Table 6.** Reduced-data 200-way retrieval at 25% and 100% training-concept coverage. Values are ten-subject means (%); subject-level dispersion is not reported. ATM denotes the semantic-alignment baseline used in this experiment.

| Evaluation    | Method     | Fraction | Recall@1 ↑   | Recall@5 ↑   | Recall@10 ↑  | MRR ↑        |
|---------------|------------|----------|--------------|--------------|--------------|--------------|
| Intra-subject | ATM        | 100%     | 27.30        | 56.95        | 70.90        | 41.40        |
|               | PerceptEEG | 100%     | <b>65.85</b> | <b>91.40</b> | <b>97.05</b> | <b>77.18</b> |
|               | ATM        | 25%      | 11.10        | 30.65        | 44.30        | 21.89        |
|               | PerceptEEG | 25%      | <b>30.95</b> | <b>65.25</b> | <b>80.40</b> | <b>46.44</b> |
| Cross-subject | ATM        | 100%     | 5.35         | 18.60        | 30.00        | 13.61        |
|               | PerceptEEG | 100%     | <b>12.85</b> | <b>34.20</b> | <b>48.05</b> | <b>24.09</b> |
|               | ATM        | 25%      | 3.90         | 14.85        | 23.95        | 10.78        |
|               | PerceptEEG | 25%      | <b>10.20</b> | <b>31.10</b> | <b>44.95</b> | <b>21.18</b> |

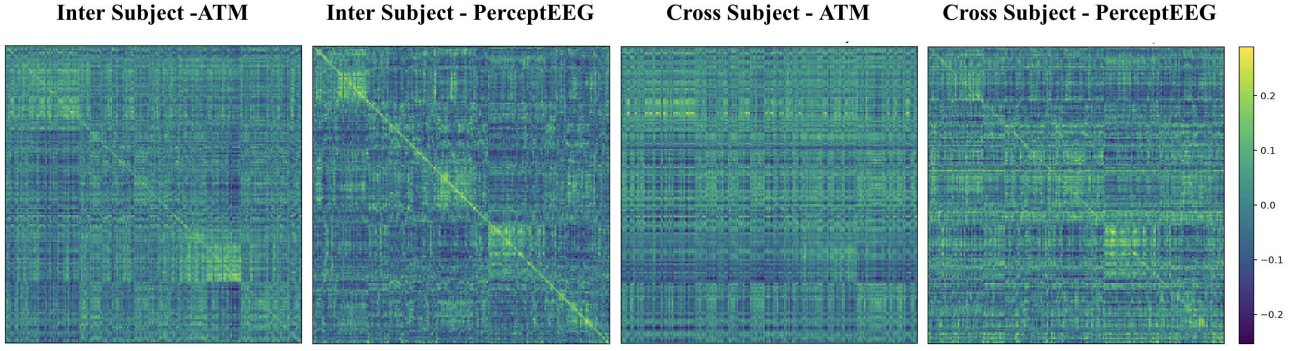
**Table 7.** Staged MVT/WSRA ablation for intra- and cross-subject 200-way retrieval. Values are mean  $\pm$  standard deviation over ten subjects; Recall and MRR are percentages. The WSRA increment is evaluated only with MVT enabled; no WSRA-only condition is included.

| Evaluation    | MVT | WSRA | Recall@1 ↑       | Recall@5 ↑       | Recall@10 ↑      | MRR (%) ↑        | Mean Rank ↓      |
|---------------|-----|------|------------------|------------------|------------------|------------------|------------------|
| Intra-subject | ✓   | ✓    | 65.85 $\pm$ 7.46 | 91.40 $\pm$ 5.10 | 97.05 $\pm$ 2.68 | 77.18 $\pm$ 6.22 | 2.37 $\pm$ 0.78  |
|               | ✓   | ×    | 64.30 $\pm$ 9.64 | 90.60 $\pm$ 4.51 | 96.25 $\pm$ 2.70 | 75.86 $\pm$ 7.34 | 2.51 $\pm$ 0.77  |
|               | ×   | ×    | 30.10 $\pm$ 5.32 | 62.40 $\pm$ 6.90 | 76.65 $\pm$ 6.44 | 45.14 $\pm$ 5.42 | 9.51 $\pm$ 2.64  |
| Cross-subject | ✓   | ✓    | 12.85 $\pm$ 4.76 | 34.20 $\pm$ 9.85 | 48.05 $\pm$ 9.05 | 24.09 $\pm$ 6.02 | 24.55 $\pm$ 6.22 |
|               | ✓   | ×    | 9.75 $\pm$ 4.62  | 30.25 $\pm$ 8.18 | 42.80 $\pm$ 7.81 | 20.62 $\pm$ 5.85 | 27.25 $\pm$ 7.56 |
|               | ×   | ×    | 6.70 $\pm$ 2.58  | 22.15 $\pm$ 5.89 | 34.35 $\pm$ 7.39 | 15.85 $\pm$ 3.71 | 36.46 $\pm$ 6.30 |

and 90.9% of Recall@5 in the cross-subject setting. These results support improved mean sample efficiency at the evaluated 25% operating point. Across the broader curve in Fig. 4, PerceptEEG remains above ATM for both metrics under intra-subject evaluation and for cross-subject Recall@5; its cross-subject Recall@1 is tied with ATM at 5% and higher at the remaining fractions.

## 4.5 Ablation Study

Table 7 adds MVT and WSRA sequentially to a controlled semantic-alignment baseline under the PerceptEEG training pipeline. The baseline is therefore an experiment-specific control rather than the published ATM entry in Table 5. In this staged ablation, adding MVT produces the largest measured increment: intra-subject Recall@1 rises from 30.10% to 64.30%, while cross-subject Recall@1 rises from 6.70% to 9.75%. Adding WSRA increases Recall@1 further to 65.85% and 12.85%, respectively. The absolute WSRA gain is modest in the intra-subject setting but larger under LOSO transfer (+3.10 points, or 31.8% relative to MVT alone). Cross-subject MRR also increases from 20.62% to 24.09%.



**Fig. 5.** Sub-01 EEG–image cosine-similarity matrices in each model’s native retrieval space. Rows are visual targets, columns are EEG queries, and correct pairs lie on the diagonal. ATM uses final CLIP features; PerceptEEG uses MVT.

## 4.6 Representation Analysis

We analyze the native retrieval spaces of the two models on Sub-01 using the full training set. RSA [20]  $\rho$  is the Spearman correlation between the upper triangles of the EEG–EEG and visual–target–visual–target cosine-similarity matrices. We additionally report the mean cosine similarity of matched pairs, the mean over all unmatched pairs, and their difference. Because ATM and PerceptEEG use different visual targets, this analysis characterizes each model’s deployed retrieval geometry; it is not a controlled comparison in a shared target space.

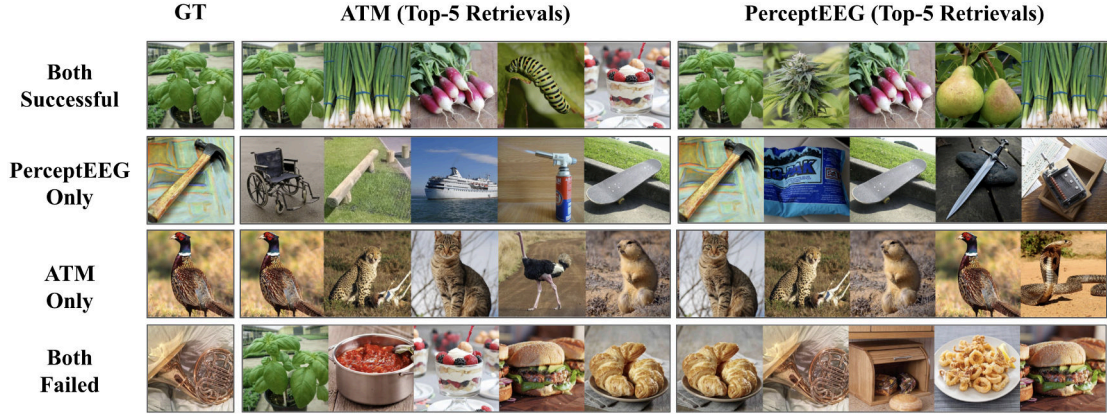
Table 8 shows higher native-space alignment for PerceptEEG in both protocols. Under intra-subject evaluation, RSA increases from 0.1114 to 0.3370, and the matched–unmatched gap increases from 0.0586 to 0.2739. Under cross-subject LOSO, RSA increases from 0.0773 to 0.2360, while the gap increases from 0.0394 to 0.1152. The matrices in Fig. 5 are consistent with these statistics. PerceptEEG shows a clearer diagonal and lower off-diagonal responses, provide a geometric account of the retrieval improvement.

**Table 8.** Full-data native-space representation analysis for Sub-01. ATM uses final CLIP features and PerceptEEG uses MVT. Bold indicates the preferred value within each setting; lower unmatched cosine is better.

| Setting       | Model             | Target     | RSA $\rho \uparrow$ | Matched $\uparrow$ | Unmatched $\downarrow$ | Gap $\uparrow$ |
|---------------|-------------------|------------|---------------------|--------------------|------------------------|----------------|
| Intra-subject | ATM               | Final CLIP | 0.1114              | 0.0854             | 0.0268                 | 0.0586         |
|               | <b>PerceptEEG</b> | MVT        | <b>0.3370</b>       | <b>0.2832</b>      | <b>0.0093</b>          | <b>0.2739</b>  |
| Cross-subject | ATM               | Final CLIP | 0.0773              | 0.0517             | 0.0123                 | 0.0394         |
|               | <b>PerceptEEG</b> | MVT        | <b>0.2360</b>       | <b>0.1160</b>      | <b>0.0009</b>          | <b>0.1152</b>  |

## 4.7 Qualitative Results

Figure 6 compares selected Sub-01 results from ATM and PerceptEEG across four outcomes: *Both Successful*, *PerceptEEG Only*, *ATM Only*, and *Both Failed*. PerceptEEG more consistently retrieves



**Fig. 6.** Selected Sub-01 examples showing the ground-truth image and ranked top-five retrievals from ATM and PerceptEEG, ordered from left to right.

**Table 9.** Exploratory extensions under 200-way retrieval. Values are mean  $\pm$  standard deviation across the indicated participants; Recall and MRR are percentages. The cross-subject PerceptEEG-Semantic result includes eight participants and is not used for a matched comparison.

| Setting       | Model               | $n$ | Recall@1 $\uparrow$ | Recall@5 $\uparrow$ | MRR $\uparrow$   |
|---------------|---------------------|-----|---------------------|---------------------|------------------|
| Intra-subject | PerceptEEG          | 10  | 65.85 $\pm$ 7.46    | 91.40 $\pm$ 5.10    | 77.18 $\pm$ 6.22 |
|               | PerceptEEG-Semantic | 10  | 59.90 $\pm$ 9.13    | 88.65 $\pm$ 4.61    | 72.51 $\pm$ 7.10 |
|               | PerceptEEG-Cross    | 10  | 64.90 $\pm$ 9.29    | 92.10 $\pm$ 4.94    | 76.63 $\pm$ 7.23 |
| Cross-subject | PerceptEEG          | 10  | 12.85 $\pm$ 4.76    | 34.20 $\pm$ 9.85    | 24.09 $\pm$ 6.02 |
|               | PerceptEEG-Semantic | 8   | 12.94 $\pm$ 3.54    | 33.94 $\pm$ 8.11    | 24.08 $\pm$ 4.56 |
|               | PerceptEEG-Cross    | 10  | 13.95 $\pm$ 4.10    | 35.80 $\pm$ 8.47    | 25.54 $\pm$ 5.44 |

images sharing perceptual attributes such as color and shape, as illustrated by the vegetation and hammer examples. Its failures tend to occur among images with more complex color or texture patterns and similar shapes. Notably, even when the top-ranked result is incorrect, PerceptEEG often retrieves the correct image within the top five, suggesting that its perceptually aligned representations capture useful visual similarity but provide limited semantic information for fine-grained ranking.

## 4.8 Exploratory Extensions

We further evaluate two exploratory extensions of the main framework. These experiments test whether late semantic fusion or an explicitly cross-participant residual can address the remaining errors.

**PerceptEEG-Semantic.** To test whether final-layer semantic information complements perceptual alignment, we retain the ATM-WSRA encoder and MVT target, and add a final-layer CLIP branch, a bounded semantic residual for score fusion, and auxiliary semantic and fused-alignment objectives. As shown in Table 9, this compound extension lowers intra-subject Recall@1 from 65.85% to 59.90%

and Recall@5 from 91.40% to 88.65%. It therefore provides no advantage under the evaluated design, but does not establish that semantic information is intrinsically detrimental.

**PerceptEEG-Cross.** To test whether explicitly modeling population-shared structure improves transfer, we add a low-rank residual that brings same-stimulus representations from different source participants closer. The branch is used only for the cross-subject route. It increases cross-subject Recall@1 from 12.85% to 13.95% and Recall@5 from 34.20% to 35.80%. These modest, single-seed gains are preliminary rather than statistically established.

## 5 Discussion

### 5.1 Intra-Subject Retrieval

The intra-subject results identify visual-target compatibility as a major factor in PerceptEEG’s improvement. The diagnostic study finds perceptual attributes more linearly recoverable from EEG than final-layer semantics and observes the strongest sampled EEG–CLIP correspondence at an intermediate depth. This pattern is consistent with evidence that artificial visual hierarchies progress from lower-level to increasingly abstract representations and exhibit time-resolved correspondence with cortical processing [3]. MVT operationalizes this observation by aligning EEG with a compact intermediate target. The diagnostics motivate this choice, although they do not uniquely determine the MVT architecture.

**Contributions of MVT and WSRA.** The staged ablation assigns the largest measured gain to MVT: replacing the semantic control raises intra-subject Recall@1 from 30.10% to 64.30%, whereas adding WSRA raises it further to 65.85%. WSRA thus provides a smaller, conditional gain through bounded sensor–time refinement. The Sub-01 native-space analysis is consistent with this result, showing higher RSA and a larger matched–unmatched similarity gap for PerceptEEG. Together, the evidence supports distinct roles for a compatible alignment target and complementary EEG refinement; it does not establish an independent WSRA main effect because the WSRA-only condition was not evaluated.

**Reintroducing Semantic Information.** PerceptEEG-Semantic tests whether final-layer CLIP information can complement perceptual alignment. Its lower intra-subject Recall@1, Recall@5, and MRR in Table 9 show that the evaluated late-fusion design offers no clear benefit.

## 5.2 Cross-Subject Retrieval

Cross-subject transfer remains the principal limitation. PerceptEEG achieves  $12.85 \pm 4.76\%$  Recall@1 and  $34.20 \pm 9.85\%$  Recall@5, but remains 6.19 and 11.70 percentage points below NeuroBridge, respectively. The lower semantic recoverability observed under LOSO evaluation illustrates the difficulty of preserving stimulus-related information across participants, but does not establish the cause of the retrieval gap. A more compatible visual target therefore improves alignment without eliminating participant-dependent distribution shift. Prior cross-subject BCI research similarly suggests that explicit distribution alignment or source-domain selection can be necessary for transfer [40].

With MVT fixed, WSRA raises cross-subject Recall@1 from 9.75% to 12.85%, compared with 64.30% to 65.85% intra-subject. This larger conditional increment suggests that structured sensor-time refinement may be particularly useful under participant shift, although the incomplete factorial ablation precludes a component-level causal claim.

**Cross-Subject Residual Adaptation.** PerceptEEG-Cross further tests explicit population-shared modeling. It increases cross-subject Recall@1 to  $13.95 \pm 4.10\%$  and Recall@5 to  $35.80 \pm 8.47\%$ . Because the gains are modest and each configuration uses a single seed, they constitute preliminary evidence rather than a statistically established improvement. The cross-subject route is disabled during intra-subject evaluation; consequently, the small intra-subject differences in Table 9 may reflect optimization variability and should not be interpreted as a specialization cost.

## 5.3 Low-Data Retrieval

PerceptEEG remains effective when training-stimulus diversity is reduced. At 25% concept coverage, it reaches 10.20% cross-subject Recall@1 and 31.10% Recall@5, retaining 79.4% and 90.9% of its full-data performance while outperforming ATM on both metrics. A plausible explanation is that the compact perceptual target simplifies cross-modal mapping while the bounded residual limits destructive changes to the pretrained EEG feature; the present experiments do not isolate these mechanisms. Self-supervised EEG pretraining provides a complementary route to data efficiency through more transferable representations [41].

## 6 Limitations and Future Work

**Evaluation Strength and External Validity.** Each configuration uses one training seed, so the reported standard deviations measure participant variability rather than optimization uncertainty. Small

method differences therefore require multi-seed evaluation and paired significance testing. The staged ablation also omits the WSRA-only condition, and the reduced-data curve lacks dispersion estimates. Future evaluation should complete the  $2 \times 2$  component design and report uncertainty at every data fraction. In addition, the study covers ten THINGS-EEG2 participants in a controlled natural-object paradigm. Validation on larger and more diverse EEG datasets, and extension to temporally resolved modalities such as MEG, are needed to assess external validity across populations, acquisition systems, and neural-signal regimes.

**Diagnostic Scope and Target Selection.** The depth analysis samples four blocks from one OpenCLIP ViT-H/14 model and selects Block 14 using Sub-01 before applying it to all participants. Because the same backbone constructs MVT, the experiment neither provides independent cross-encoder validation nor establishes a universally optimal depth. Future work should repeat selection across participants and compare CLIP with handcrafted perceptual targets, supervised encoders, and self-supervised representations such as DINO [42]. The sensor–time masking analysis measures sensitivity to retained inputs rather than additive regional contributions or neural source localization, and the RSA analysis is likewise limited to Sub-01.

**Architectural Generality.** WSRA is evaluated only with ATM and uses fixed branch weights and a fixed residual bound. Regularized dynamic gating could adapt the sensor–time views to individual trials or participants, while tests across diverse EEG encoders would determine whether the gain reflects a general advantage of residual adaptation or an ATM-specific interaction.

**Semantic Integration and Cross-Subject Generalization.** The semantic and cross-subject extensions are preliminary. The semantic variant changes several components simultaneously and tests only one late-fusion strategy; the cross-subject variant yields small single-seed gains. Future work should investigate reliability-aware semantic gating, stronger population-scale self-supervision, and explicit subject-invariant objectives under matched seeds and participant coverage. These directions may help retain useful high-level information without overwhelming perceptual evidence and reduce the dependence of EEG representations on participant identity.

## 7 Conclusions

This dissertation examined a basic assumption in EEG-to-image retrieval: whether visually rich representations are necessarily appropriate alignment targets for EEG. A three-part diagnostic study characterized what information is recoverable from EEG, when and where that information is most

useful, and which sampled visual depth is most compatible with the signal. The results show a systematic imbalance: perceptual attributes are more recoverable than final-layer semantics, performance is most sensitive to posterior activity in the 100–300 ms interval, and middle-layer alignment provides the strongest sampled EEG–CLIP correspondence. These findings diagnose target compatibility rather than providing neural source localization or a universal optimal visual layer.

Motivated by this evidence, PerceptEEG combines a Mid-level Visual Target (MVT) with a Windowed Spatiotemporal Residual Adapter (WSRA). On THINGS-EEG2, it achieves  $65.85 \pm 7.46\%$  Recall@1 and  $91.40 \pm 5.10\%$  Recall@5 in intra-subject retrieval, the highest mean performance among the included methods. Under cross-subject LOSO evaluation, it reaches  $12.85 \pm 4.76\%$  and  $34.20 \pm 9.85\%$ , improving on ATM but remaining below NeuroBridge. With 25% of the training concepts, it retains 79.4% of full-data Recall@1 and 90.9% of Recall@5 in the cross-subject setting. The staged ablation identifies MVT as the principal measured contributor and supports WSRA only as a conditional refinement in the evaluated design.

Overall, the study shows that aligning EEG with a compatible visual representation can be more effective than matching the richest available image embedding. It also highlights robust cross-subject transfer as a central unresolved challenge. Future work should investigate broader datasets, complete factorial and multi-seed evaluation, cross-encoder validation of visual targets, and more effective approaches to semantic and subject-invariant modeling, thereby assessing the generalizability of the framework beyond the present setting.

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